STAT 387 **HOMEWORK # 2** Winter 2026 INTRODUCTION TO STATISTICAL LEARNING

DUE DATE: **January 28, 11:59 pm**

Instructions:

- *Please show all the work with proper mathematical/statistical notation and logical flow to receive full points.*
- *You are allowed to use **R** or a Scientific Calculator where required.*
- *Submit only **one** **PDF** on* CANVAS.

All questions here are taken from the Exercises pages in the book for Chapter 3 (**pages 121 – 128**): Introduction to Statistical Learning with Applications in R

Conceptual

1. 2) KNN Classifier and KNN regression both use a positive integer K and a prediction point of x_0, and identify the set of K points in the training data set that are closest to x_0, which forms the K nearest neighbors of x_0. This set could be represented as n_0.

The key differences lie in how they utilize this set of neighbors. The KNN classifier estimates the conditional probabilities of the class and assigns x_0 to the class with the largest probability. Classification is determined by a majority vote, with ties broken at random. If there are ties for the K-th nearest neighbor, then all candidates are typically included in the vote. Flexibility is based on the neighbor count K; as it increases, flexibility decreases, which implies that bias will increase and variance will decrease.

KNN Regression will first identifying the neighbors just like KNN classifier, but it uses mean as the average response in n_0. It is a local,

nonparametric method that uses K to control flexibility which also affects the bias-variance trade off, similar to the classifier method. When K=1, assuming we will also have no ties, then we have $y_i = \hat{y}_i$, meaning the training error is zero. While KNN regression is simple and does not require a priori assumptions about the shape of f, there are no associated tests for its validity and this method will break down when *p* is large or *n* is not large.

2. 3 (a, b)

a) For a fixed value of IQ and GPA, Males will earn more on average provided the GPA is high enough, with that crossover point being around 3.5 GPA.

b)

$$Salary = 50 + 20(4) + 0.07(110) + 35(1) + 0.01(440) - 10(4)$$

$$Salary = 137.1$$

The salary for a female employee with a 4.0 GPA and 110 IQ is $137,100

Applied

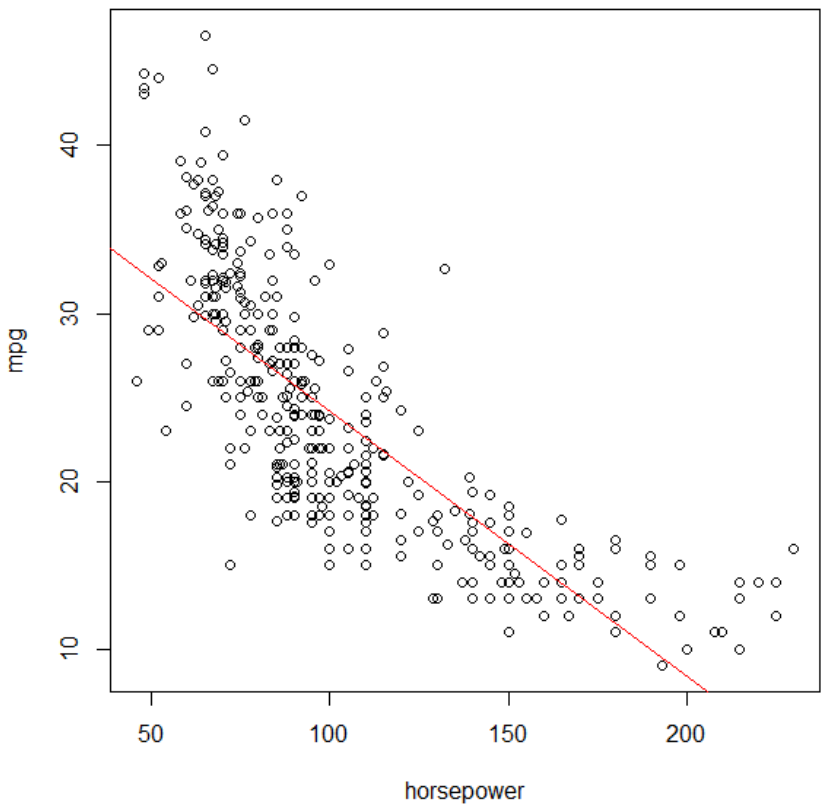
3. 8 (a, b, c)

a)i) Since our p value is very small, we can reject the null hypothesis and assume there is a relationship between the predictor and the response in the horsepower and mpg linear model.

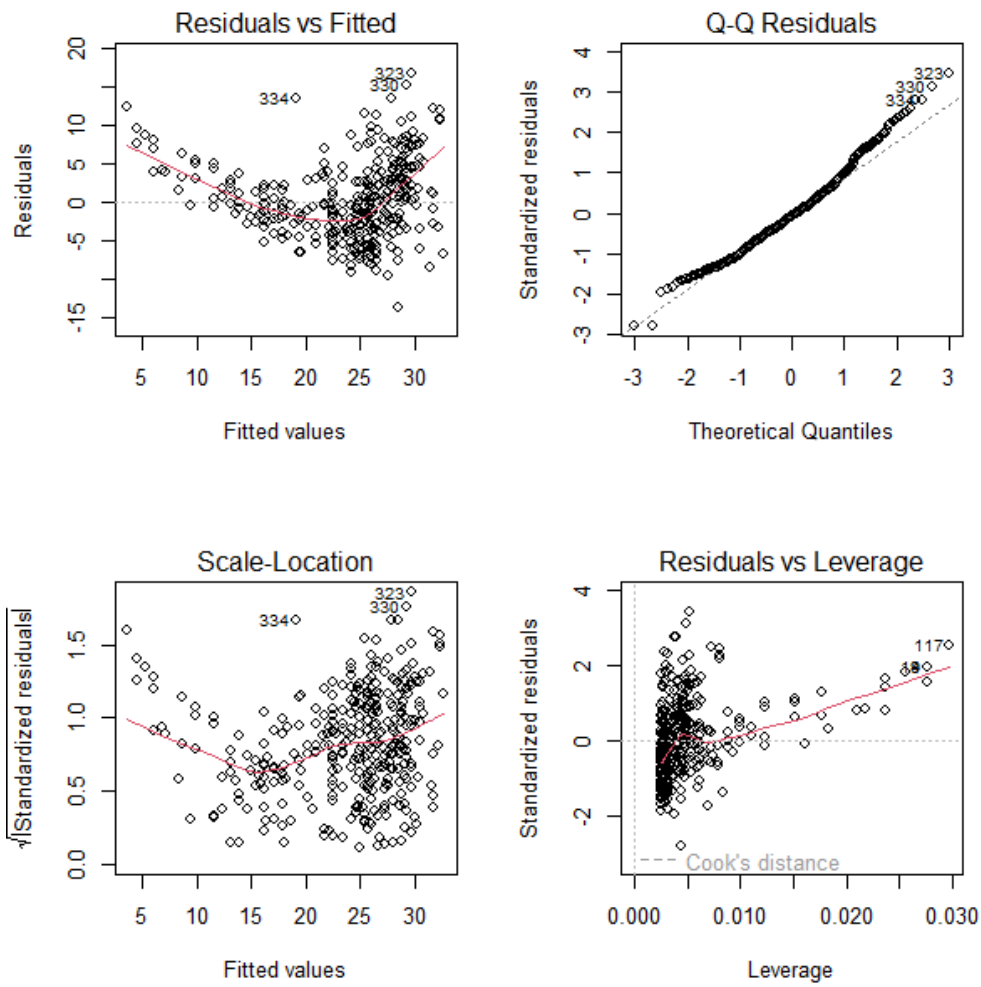
ii) The relationship is decent but could be stronger. We have an r-squared value of 0.6059 which is not very high and indicates that only 60% of the variance in mileage is explained by engine horsepower.

iii) The relationship is negative, generally meaning that as horsepower increases miles per gallon will decrease.

iv) The predicted MPG associated with a horsepower of 98 is 24.46708. The confidence interval is (23.97308, 2496108) and the prediction interval is (14.8094, 34.12479)



b)



c)

The residual vs fitted plot shows a distinct u-shaped curved pattern rather than a straight horizontal line. This indicates a non-linear relationship between horsepower and mpg and further suggests that a simple linear regression might not be the best fit.

The Q-Q deviates quite a lot on the right side and a little bit less on the left. Normality might be violated in this model.

The scale-location plot shows an upwards trend with a lot of fanning of points at either end of the plot, suggesting that homoscedasticity is

violated in this model.
Residuals vs Leverage shows most of the points towards the left side of the plot, with the red line curving upwards as the graph progresses, suggesting a few noticeable outliers in the model.

4. 10 (a, b, c, d, e, f)

a)

```
Call:
lm(formula = Sales ~ Price + Urban + US)

Residuals:
    Min      1Q  Median      3Q     Max 
-6.9206 -1.6220 -0.0564  1.5786  7.0581 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 13.043469   0.651012  20.036  < 2e-16 ***
Price       -0.054459   0.005242 -10.389  < 2e-16 ***
UrbanYes    -0.021916   0.271650  -0.081    0.936    
USYes        1.200573   0.259042   4.635 4.86e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.472 on 396 degrees of freedom
Multiple R-squared:  0.2393,	Adjusted R-squared:  0.2335 
F-statistic: 41.52 on 3 and 396 DF,  p-value: < 2.2e-16
```

b)Price: For every unit increase in price, sales will decrease by 0.054459, assuming that other variables remain constant.
UrbanYes: A store in an urban location sells 0.0219 less units than a store in a non-urban location.
USYes: A store based in the United States sells 1.200573 more units on average than a store located outside the US.

c)

$$Price = 13.0435 - 0.0545(Price) - 0.0219(UrbanYes) + 1.200573(USYes) + \varepsilon$$

d) Testing H0 : βj = 0 asks whether a predictor has a statistically significant effect on sales. For our linear model we can reject the null

hypothesis for Price and US since they both have a statistical effect on sales. We fail to reject the null hypothesis for Urban since its p-value is large and this suggests that Urban does not have statistical significance on sales.

```
Call:
lm(formula = Sales ~ Price + US)

Residuals:
    Min      1Q  Median      3Q     Max 
-6.9269 -1.6286 -0.0574  1.5766  7.0515 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 13.03079    0.63098  20.652  < 2e-16 ***
Price       -0.05448    0.00523 -10.416  < 2e-16 ***
USYes        1.19964    0.25846   4.641 4.71e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.469 on 397 degrees of freedom
Multiple R-squared:  0.2393,	Adjusted R-squared:  0.2354 
F-statistic: 62.43 on 2 and 397 DF,  p-value: < 2.2e-16

>
```

e)

f) The models in a and e have almost the same R-squared score, meaning they both account for about 24% of the variance in sales. Since the R-squared values are so similar and since UrbanYes had such a high p-value that made it fail to reject the null hypothesis, we would benefit more from the simpler model in e that only includes statistically significant predictors.

5. 13 (a, b, c, d, e)

a)

```
> x<-rnorm(100)
> head(x)
[1]  0.90193791  0.44943321 -1.04444540  0.45388109 -2.54664008 -0.03673285
>
```

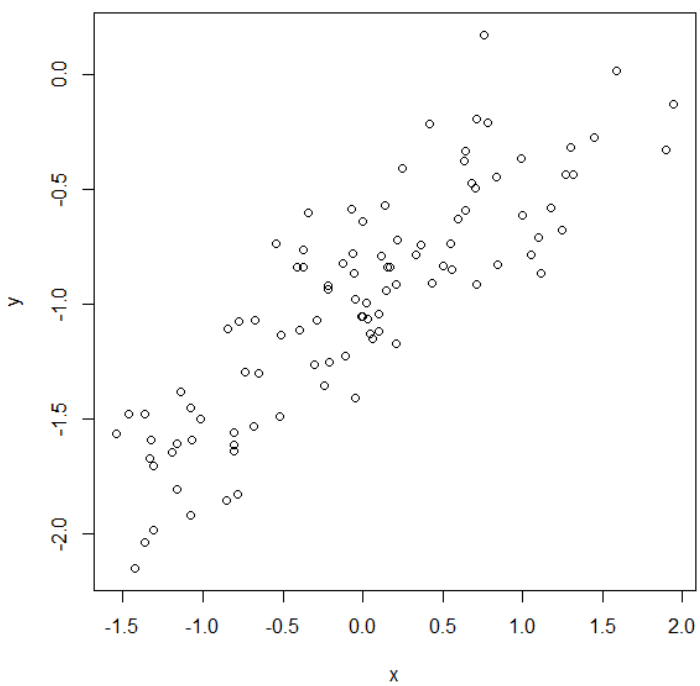
b)

```
> eps<-rnorm(100,0,0.25)
> head(eps)
[1]  0.21005191  0.21663312 -0.07207834  0.29538323 -0.11129262  0.44235646
```

c)Since x and eps have 100 observations each, and the resulting vector will also have a length of 100, since r performs vector arithmetic in an element wise fashion. $\beta 0 = -1$ and $\beta 1 = 0.5$

```
> head(y)
[1] -0.5743378 -1.2442514 -0.8579666 -1.4388454 -1.1556270 -1.5217884
>
```

d)The plot shows a positive linear relationship between x and y, with a strong linear relationship between x and y. There is a slight amount of homoscedasticity in the scatter plot. Some slight transformations might be necessary but they might not have a strong effect since the relationship is already strong overall.



e)

```
Call:
lm(formula = y ~ x)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.45945 -0.18482 -0.01812  0.16348  0.75942 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -0.97306    0.02586  -37.62   <2e-16 ***
x            0.50749    0.03113   16.30   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.2585 on 98 degrees of freedom
Multiple R-squared:  0.7306,	Adjusted R-squared:  0.7278 
F-statistic: 265.8 on 1 and 98 DF,  p-value: < 2.2e-16

>
```

The least squares model resulted in an estimated intercept of $\hat{\beta}_0$ =-0.97306 and a slope of $\hat{\beta}_1$=0.50749, When compared to the true population values of β_0=-1.0 and β_1=0.5, the estimates are remarkably close. The model exhibits highly significant p-values and an acceptable R-Squared value, indicating a strong fit. These results demonstrate that the linear regression successfully captured the underlying signal despite the random noise added to the data.